



Analog IC Design

CMOS Technology

MOHAMMED NASR EL-DIN

Section 3 – Lab02

Common source amplifier

Common Source Amplifier

PART 1: Sizing Chart

The technology used: 180nm CMOS, So the supply V_{DD} will be 1.8v.

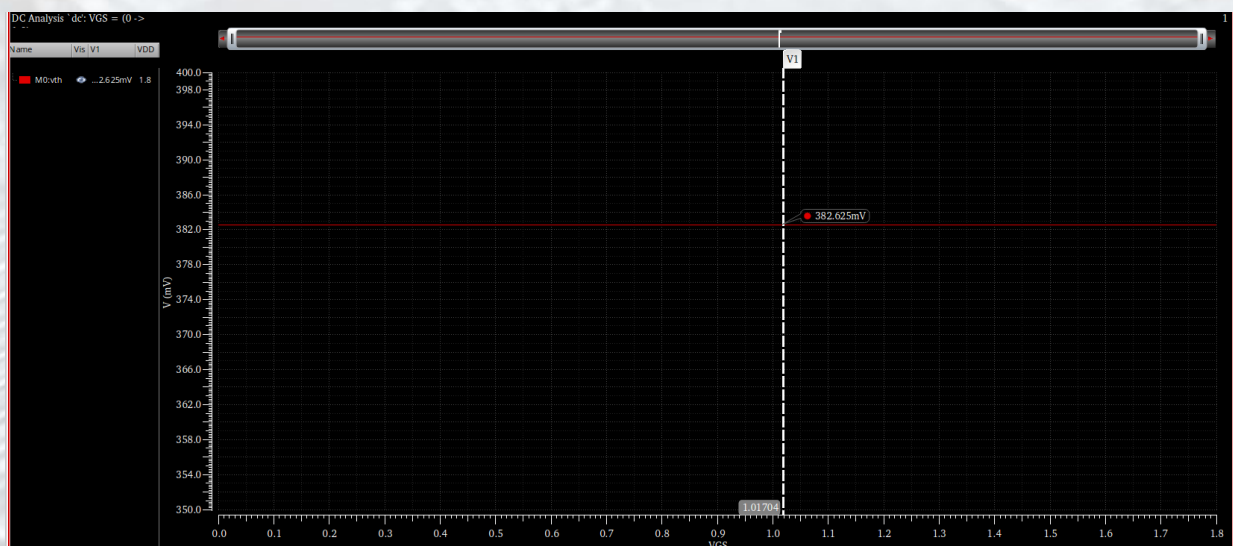
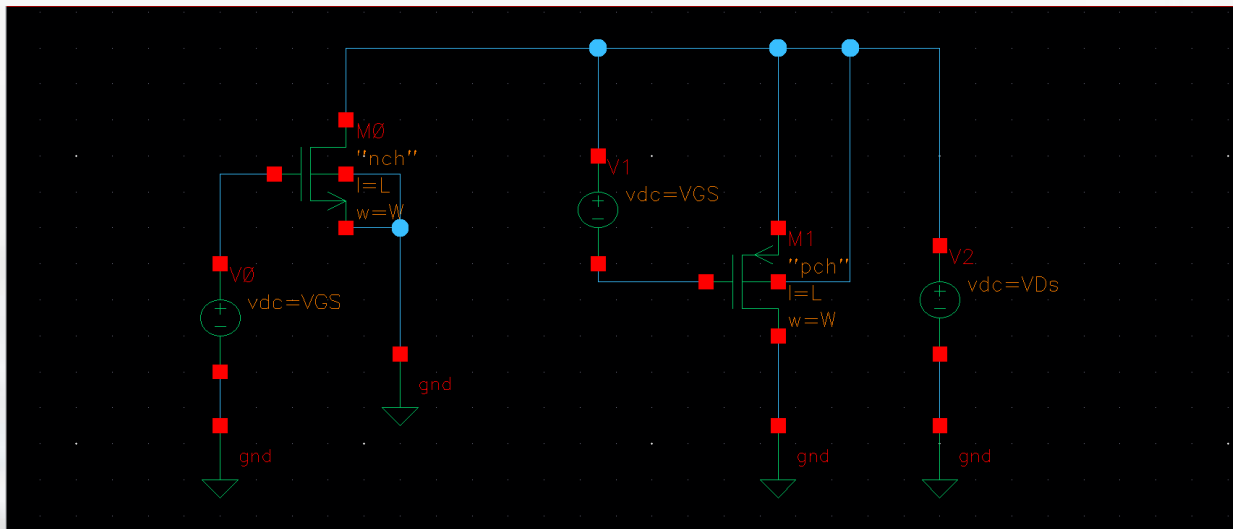
DC Gain = $A_v = -8$, Current consumption = $I_D = 100\mu A$.

$$|A_v| \approx g_m R_D = \frac{2I_D}{V_{ov}} \times R_D = \frac{2V_{RD}}{V_{ov}} \rightarrow g_m = \frac{2V_{RD}}{V_{ov} R_D} = \frac{2I_D}{V_{ov}}, \text{ Assume } L = 2\mu m.$$

$$V_{RD} = \frac{V_{DD}}{2} = \frac{1.8v}{2} = 0.9v$$

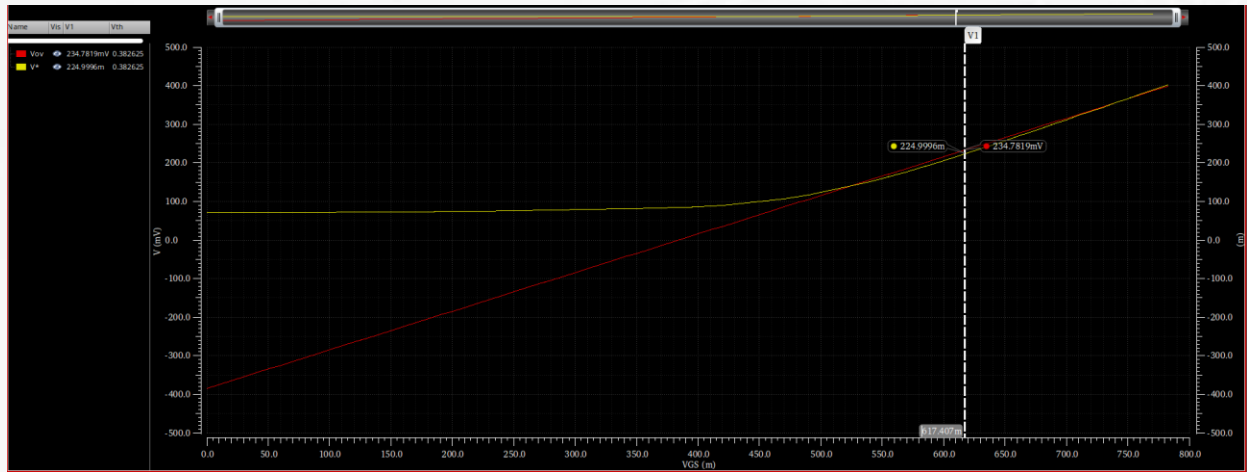
$$V_{RD} = I_D \times R_D = 0.9v \rightarrow R_D = \frac{V_{RD}}{I_D} = \frac{0.9v}{100\mu A} = 9k\Omega$$

$$|A_v| = \frac{2V_{RD}}{V^*} = 8 \rightarrow V^* = \frac{2V_{RD}}{|A_v|} = \frac{2 \times 0.9v}{8} = 225mv = V_Q^*, V_{DS} = \frac{V_{DD}}{2} = \frac{1.8v}{2} = 0.9v.$$



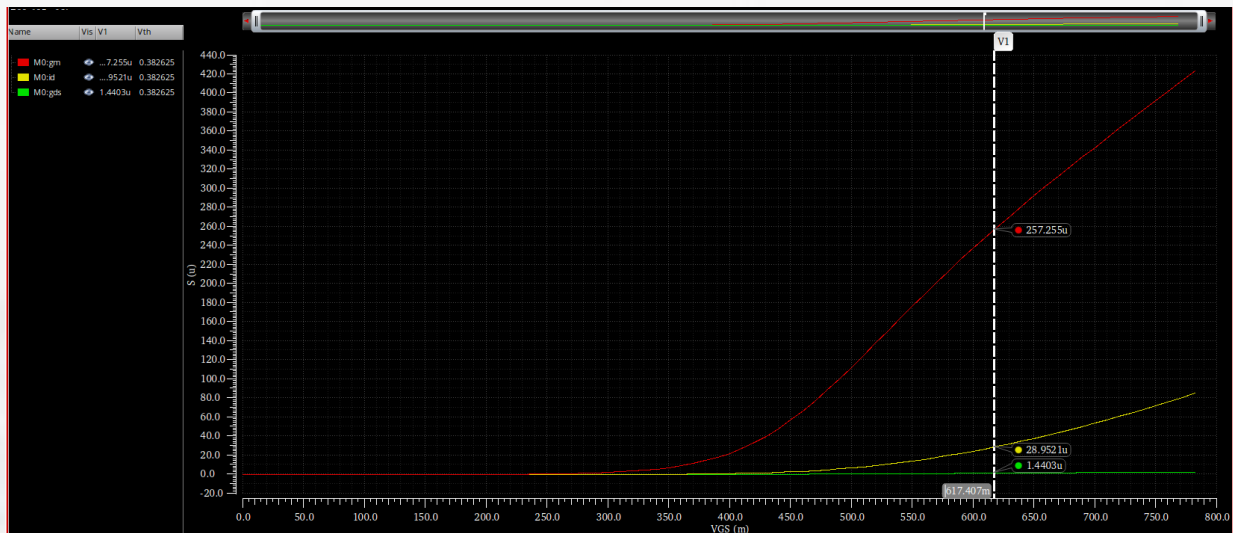
We can get V_{TH} from result browser $\rightarrow V_{TH} = 382.625mv$.

So, we will sweep V_{GS} to $V_{TH} + 0.4v \approx 0.8v$.



$$\text{Yellow } V^* = \frac{2I_D}{g_m} \quad \text{Red } V_{ov} = V_{GS} - V_{TH}, \text{ applied } w = 10\mu\text{m}$$

The corresponding V_{ovQ} and V_{GSQ} to $V_Q^* = 225\text{mV}$ are 234.78mV and 617.407mV respectively.



$$\text{Red } g_m \quad \text{Yellow } I_D \quad \text{Green } g_{ds}$$

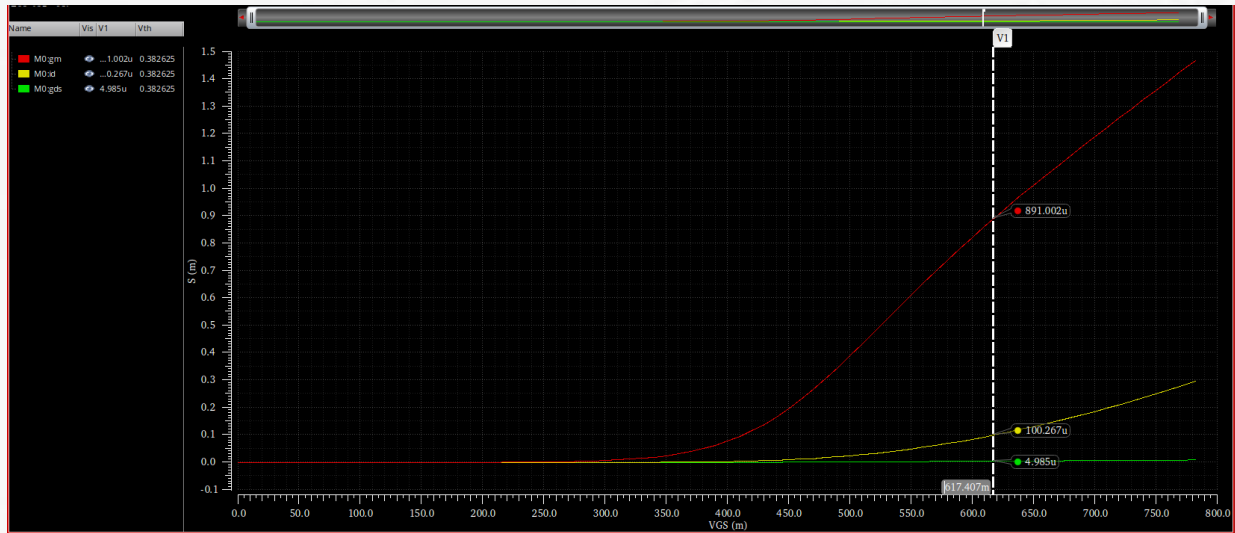
The values of I_D , g_m and g_{ds} at $V_{GSQ} = 617.407\text{mV}$ are I_{DX} , g_{mX} and g_{dsX} respectively.

$$I_{DX} \approx 29\mu\text{A}, g_{mX} \approx 257.3\mu\text{S}^{-1} \text{ and } g_{dsX} \approx 1.44\mu\text{S}^{-1}$$

$$\text{Since } I_D \propto w \rightarrow \frac{I_{D1}}{I_{D2}} = \frac{w_1}{w_2} \rightarrow w_2 = w_1 \times \frac{I_{D2}}{I_{D1}}$$

$$\text{Let } w_1 = 10\mu\text{m} \rightarrow I_{D1} = 29\mu\text{A}$$

$$\text{the width } w_2 \text{ that can provide } I_{D2} = 100\mu\text{A} \text{ given by } w_2 = w_1 \times \frac{I_{D2}}{I_{D1}} = 10\mu\text{m} \times \frac{100\mu\text{A}}{29\mu\text{A}} \approx 34.5\mu\text{m}$$



The value of I_D after setting the correct width: $100.267\mu\text{m}$

since $g_m \propto w$ and $g_{ds} \propto w \rightarrow \frac{g_{m1}}{g_{m2}} = \frac{W1}{W2}$ and $\frac{g_{ds1}}{g_{ds2}} = \frac{W1}{W2}$

If $w1 = 10\mu\text{m} \rightarrow g_{m1} \approx 257.3\mu\Omega^{-1}$ and $g_{ds1} \approx 1.44\mu\Omega^{-1}$

At $w2 = 34.5\mu\text{m}$ we can get g_{m2} and g_{ds2} by:

$$g_{m2} = g_{m1} \times \frac{w2}{w1} = 257.3\mu\Omega^{-1} \times \frac{34.5\mu\text{m}}{10\mu\text{m}} \approx 887.7\mu\Omega^{-1} \quad (\text{as shown in the graph above})$$

$$g_{ds2} = g_{ds1} \times \frac{w2}{w1} = 1.44\mu\Omega^{-1} \times \frac{34.5\mu\text{m}}{10\mu\text{m}} \approx 5\mu\Omega^{-1} \quad (\text{as shown in the graph above})$$

double check: $r_o = \frac{1}{g_{ds}} = \frac{1}{5\mu\Omega^{-1}} = 0.2M\Omega$

$$A_v = -g_m(R_D || r_o) = -8$$

$-g_m(R_D || r_o) = -887.7\mu\Omega^{-1} (9k\Omega || 0.2M\Omega) \approx -7.65 \rightarrow$ very small error due to the approximations we have made.

$\rightarrow w = 34.5\mu\text{m}$

$\rightarrow L = 2\mu\text{m}$

$\rightarrow V_{DD} = 1.8v$

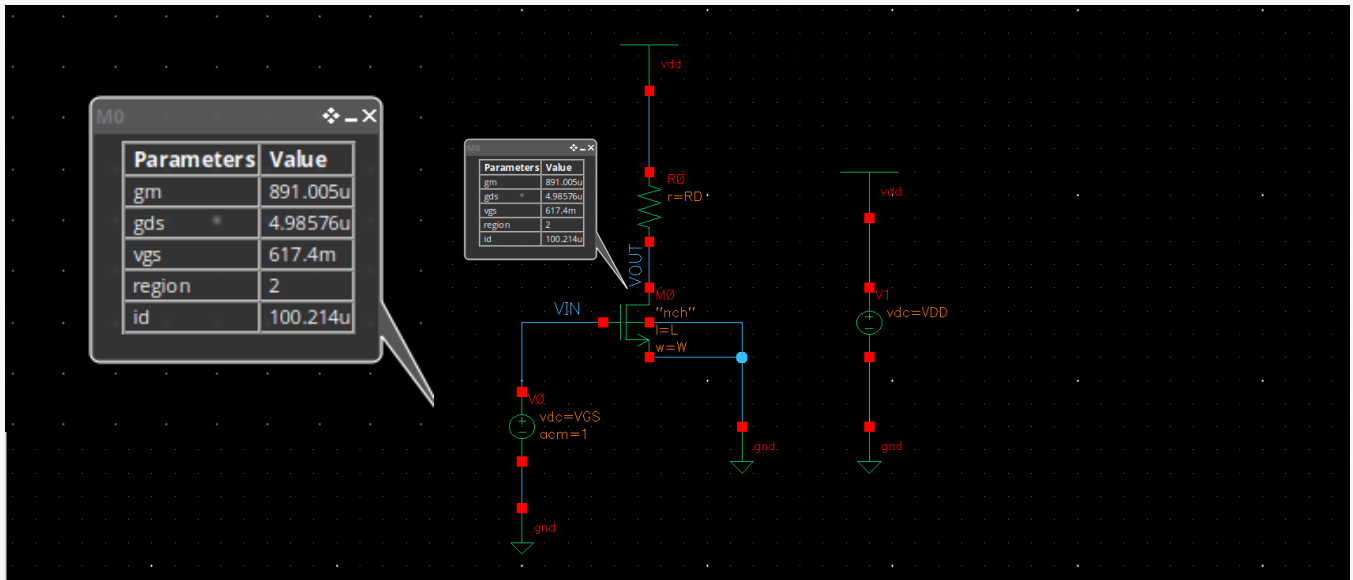
$\rightarrow V_{GS} = 617.4mv$

$\rightarrow R_D = 9k\Omega$

PART 2: CS Amplifier

1. OP and AC Analysis

1)



2) The parameter results **match** the results obtained from sizing charts.

3) $(r_o \approx 0.2M\Omega) \gg (R_D = 9K\Omega)$

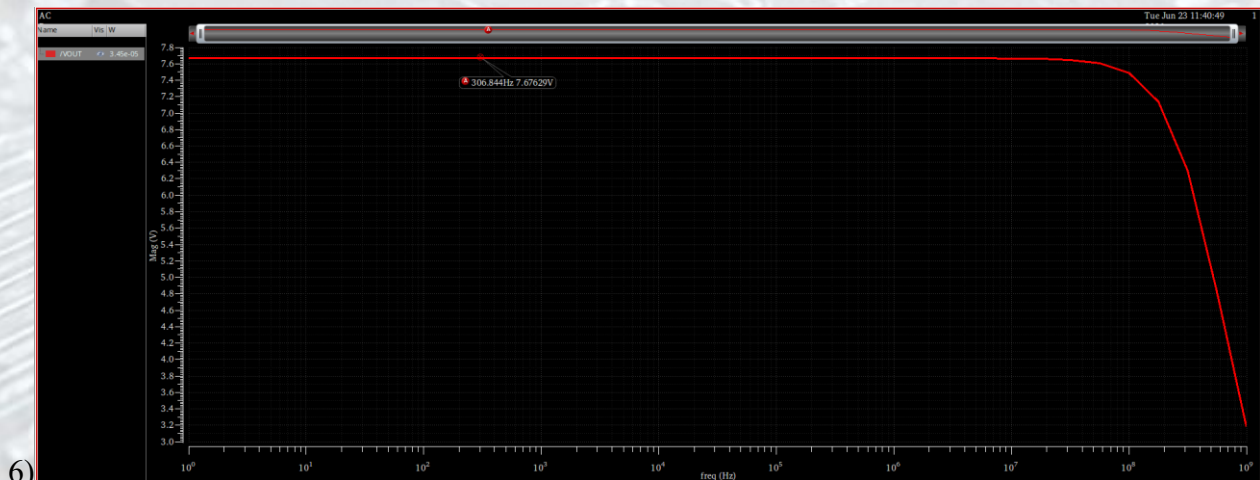
$(R_D || r_o) \approx 8.6K\Omega$, if we ignore r_o we will get only 4.65% error which is **acceptable** assumption.

For min L , r_o **decreases** due to stronger channel length modulation, so ignoring r_o becomes less accurate.

4) the intrinsic gain given by $-g_m r_o \approx -891\mu\Omega^{-1} \times 0.2M\Omega = -178.2 = A_{vi}$

5) the amplifier gain given by $-g_m (R_D || r_o) = -891\mu\Omega^{-1} \times 8.6K\Omega \approx -7.66$

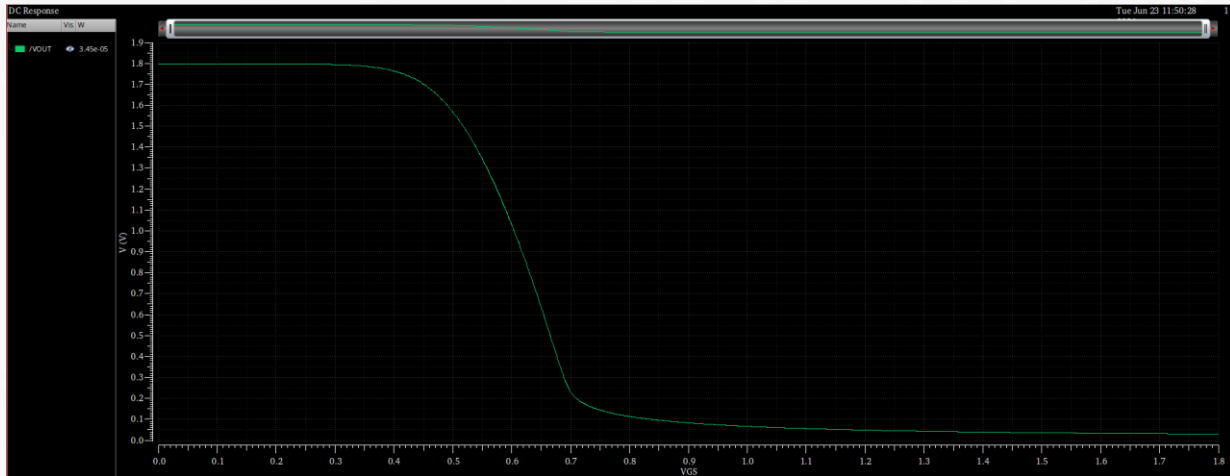
the amplifier gain is much **smaller** than intrinsic gain $A_v \ll A_{vi}$



6)

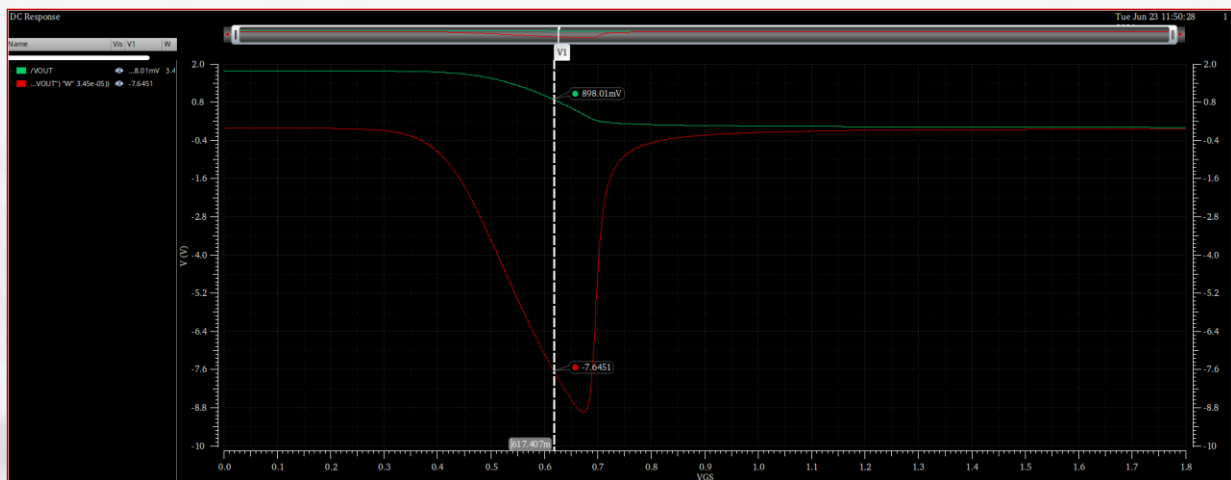
The DC gain is ≈ 7.68 which matches the spec.

2. Gain Non-Linearity



V_{out} vs V_{in} (note: $V_{in} = V_{GS}$)

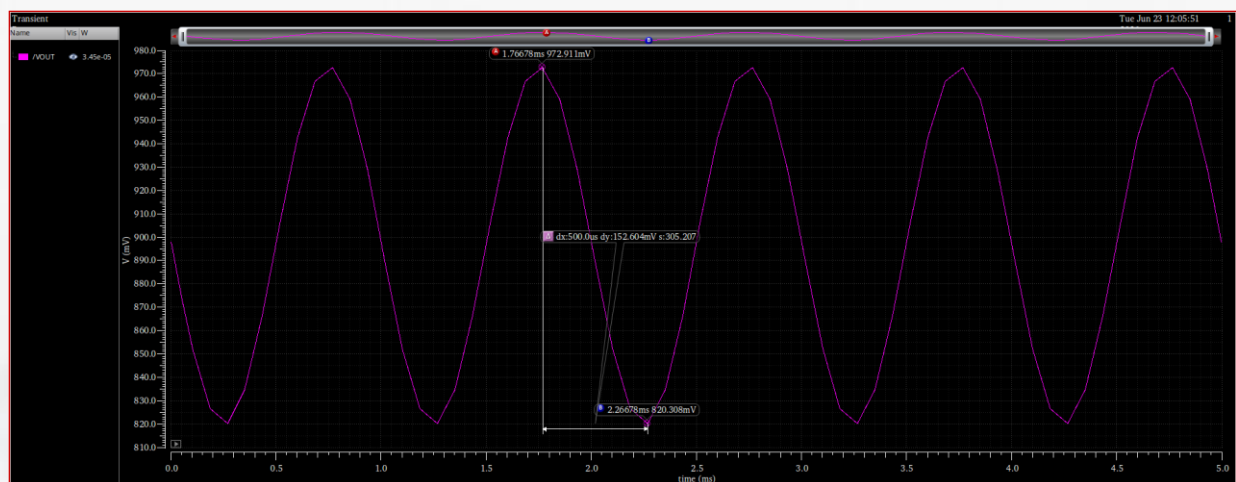
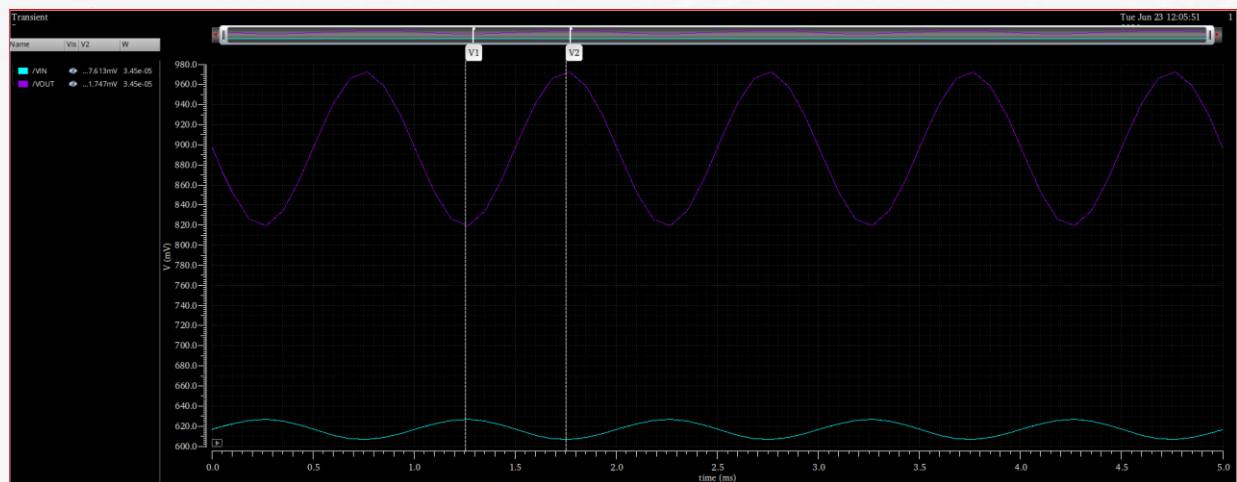
2) the relation between V_{out} and V_{in} is not linear **since** the transconductance ($g_m \propto \frac{V_{out}}{V_{in}}$) is nonlinear **due** to nonlinearity in MOSFET transfer characteristics, **but** it seems semi linear near to the bias point.



3)

($\frac{dV_{out}}{dV_{in}}$, V_{out}) vs V_{in}

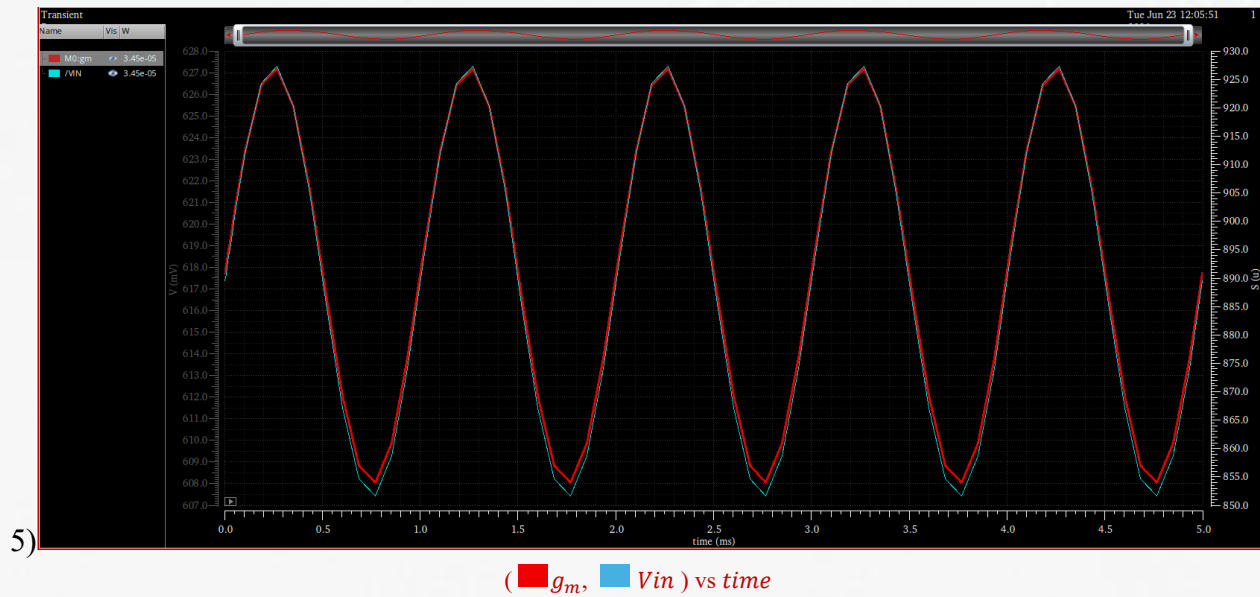
The derivative $\frac{dV_{out}}{dV_{in}}$ represents the small signal gain and it is not linear, **because** the MOSFET transfer characteristics is nonlinear.



Vout vs time

We can calculate the gain from the above waveform to ensure that the gain matches the spec.

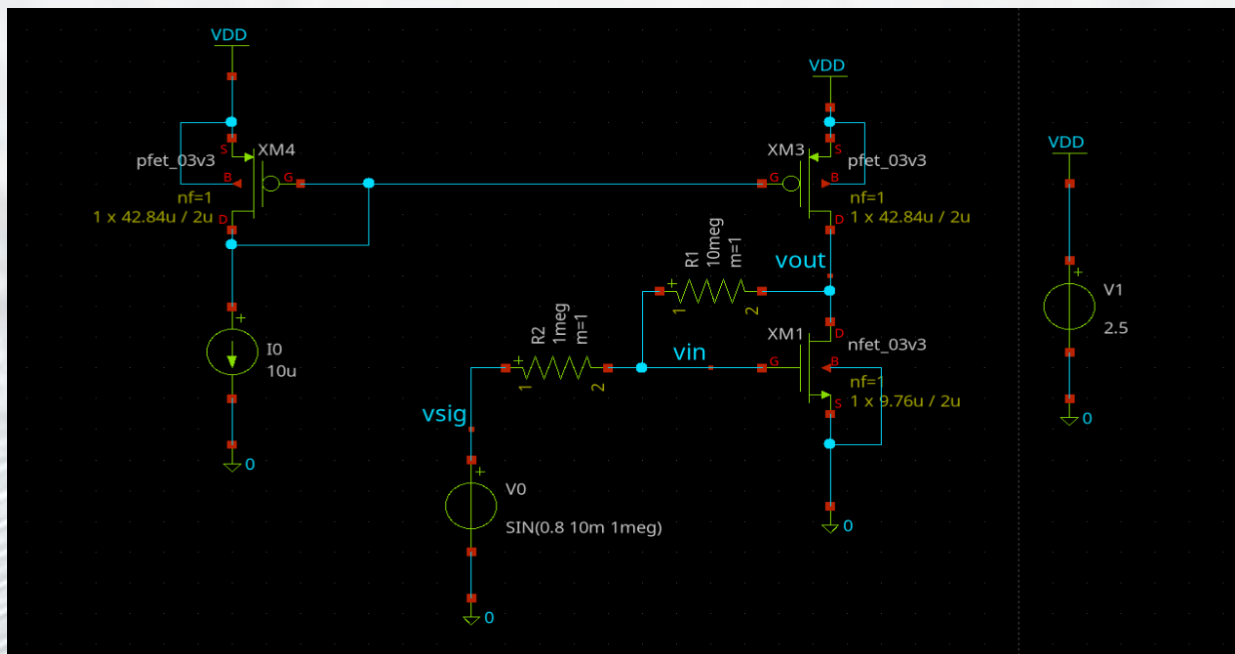
$$\text{the gain} = \frac{V_{out_{p2p}}}{V_{in_{p2p}}} = \frac{973\text{mV} - 820.3\text{mV}}{-20\text{mV}} = -7.635 \text{ which matches the spec.}$$



The waveform above shows that g_m **varies** with the input V_{in} and this **because** it depends on V_{GS} . This **means** the amplifier gain ($A_v \propto g_m$) is not constant and can change with the input signal, **leading** to nonlinear behavior and distortion for large signals.

6) The amplifier is **not** linear **because** the MOSFET exhibits a **nonlinear** square-law relationship between I_D and V_{GS} . **However**, it can be approximated as linear **only** for small signals around the bias point.

Gain Linearization (-Ve Feedback)



LUT Settings

LUT: pmos_03v3

Corner: TT

Temp (°C): 27.0

Frequency: 1

ID: 10u

Vstar: 200m

L: 2u

VDS: 1

VSB: 0

Stack: 1

Results:

Name	TT-27.0
4 W	42.84u

PMOS sizing

LUT Settings

LUT: nmos_03v3

Corner: TT

Temp (°C): 27.0

Frequency: 1

ID: 10u

Vstar: 200m

L: 2u

VDS: 2.5

VSB: 0

Stack: 1

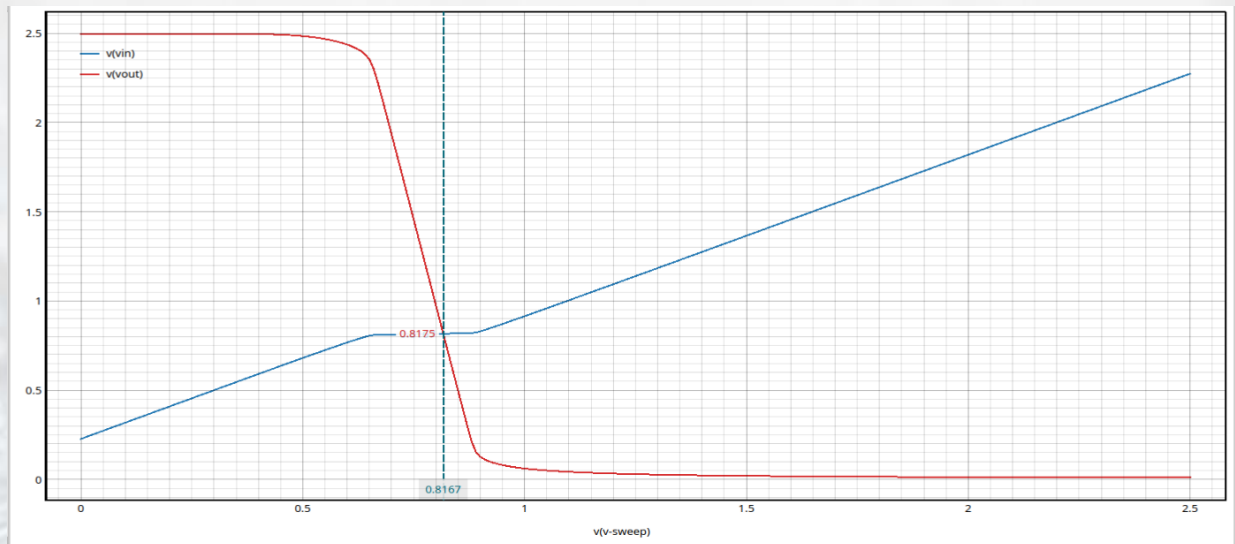
Results:

Name	TT-27.0
4 W	9.76u
5 VDB	2.5
6 VGB	816.2m
7 VGS	816.2m

NMOS sizing

$$V_{GS} = 816.2\text{mV}$$

$$R_f = 10\text{M}\Omega, \frac{R_f}{R_{in}} = |A_v| = 10 \rightarrow R_{in} = 1\text{M}\Omega$$



Vout and Vin vs Vsig

At what voltage do the two curves cross? Why?

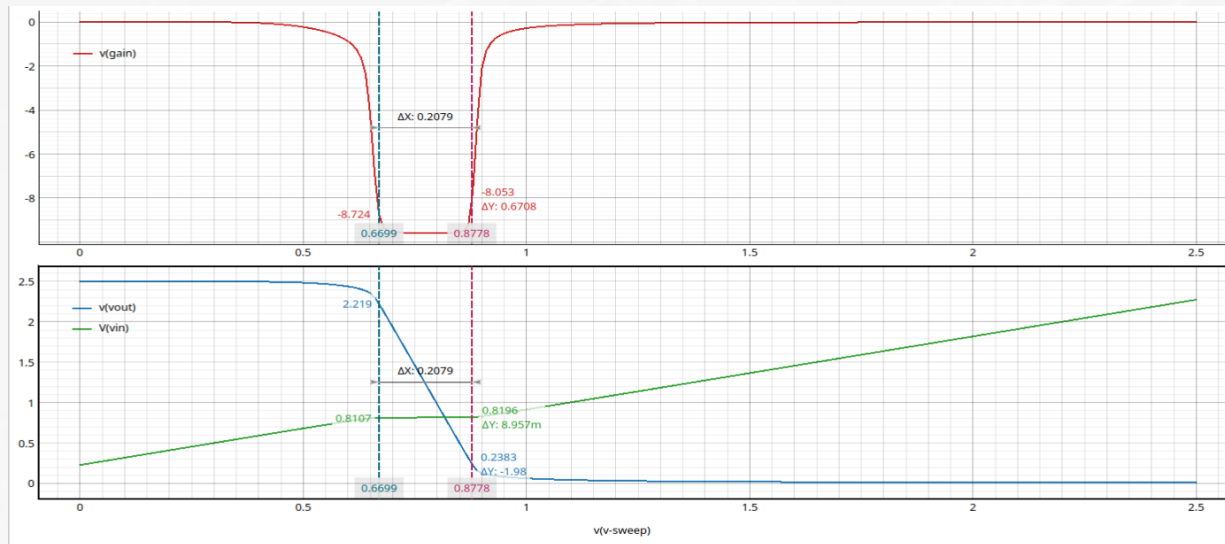
The two curves intersect at $816.7\text{mV} \approx V_{GS}$

When $V_{out} = V_{in} \rightarrow I_{R_f} = 0 \rightarrow I_{R_1} \text{ also } = 0, \rightarrow V_{R_f} = V_{R_1} = 0 \rightarrow V_{sig} = V_{out} = V_{in} = V_{GS} \approx 816\text{mV}$

Is VOUT vs VSIG linear in the operating range of the amplifier? Why?

Yes, V_{out} vs V_{sig} relation is linear in the operating range, since $V_{in} \approx \text{constant value } "V_{GS}"$

$$V_{in} - V_{out} = (V_{sig} - V_{out}) \frac{R_f}{R_1} \rightarrow \frac{R_f}{R_1} V_{sig} - \left(\frac{R_f}{R_1} - 1 \right) V_{out} = V_{in} \approx \text{constant} \rightarrow \text{linear relation}$$



Vout differentiation: gain vs Vsig

Is the gain linear (independent of the input) in the operating range of the amplifier? Why?

Since in the operating range the gain is constant (independent of the input), **yes**, the gain is linear in the operating range.

We can also calc the gain as $\frac{\Delta y}{\Delta x}$ for V_{out} curve: $= -\frac{1.98}{0.2079} \approx -9.52$

VIN is almost constant in the operating range of the amplifier. What is its value? Why?

Since the MOSFET is biased in saturation with fixed gate voltage $V_{GS} \approx 816\text{mv}$, when the input voltage reaches this value it will saturate as long as the MOSFET is still in saturation region.

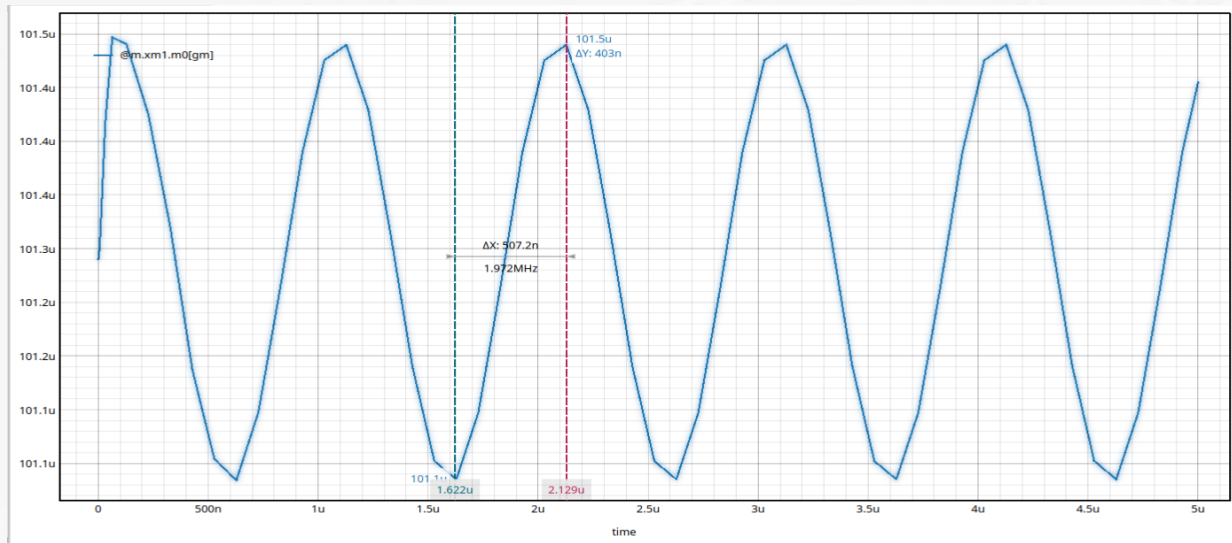
Analytically calculate the DC input range over which the gain is linear:

$$\frac{V_{DD} - 2V^*}{|A_v|} = \frac{2.5\text{v} - 2 \times 200\text{mv}}{10} = 0.21$$

The simulation input range is the range of V_{sig} at which V_{out} is linear: $\Delta x \approx 0.208$ **very small error**

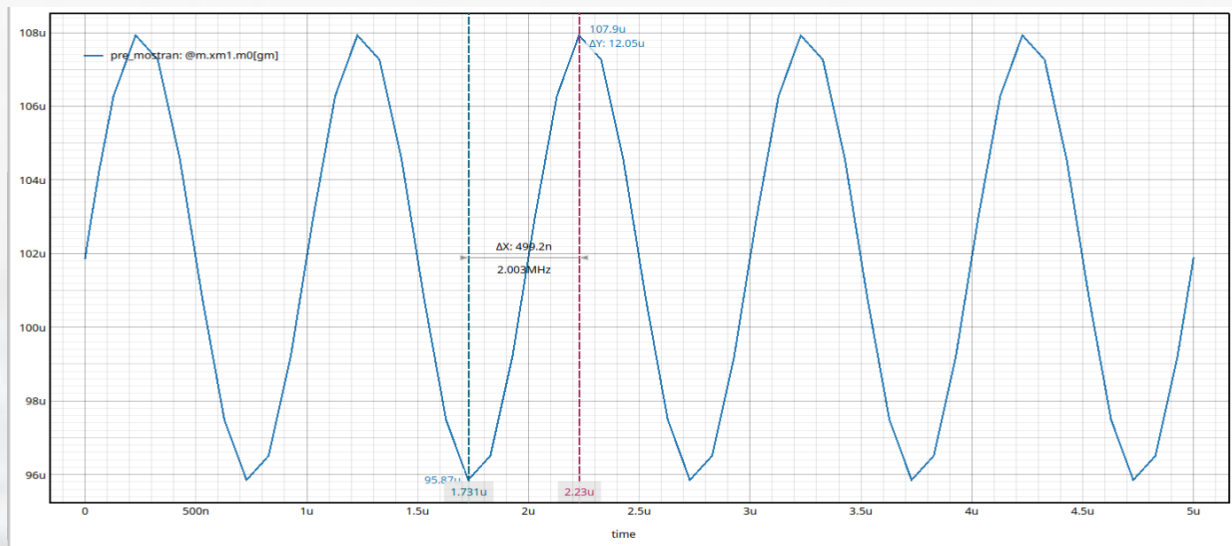
Run a transient simulation and report gm vs time. Compare this plot to the plot getting using resistive loaded.

Applying sin wave with **0.8v** DC shift to swing the input in the operating range of the MOSFET.



g_m using PMOS load

g_m has peak to peak swing of 403nS between the values (101.5μS, 101.1μS)



g_m using resistive load

g_m has peak to peak swing of 12.05μS between the values (107.9μS, 95.87μS)

In case using PMOS load: this provides **more stable** g_m variation which improves the amplifier linearity.